

Control Flow: Loop

林致宇

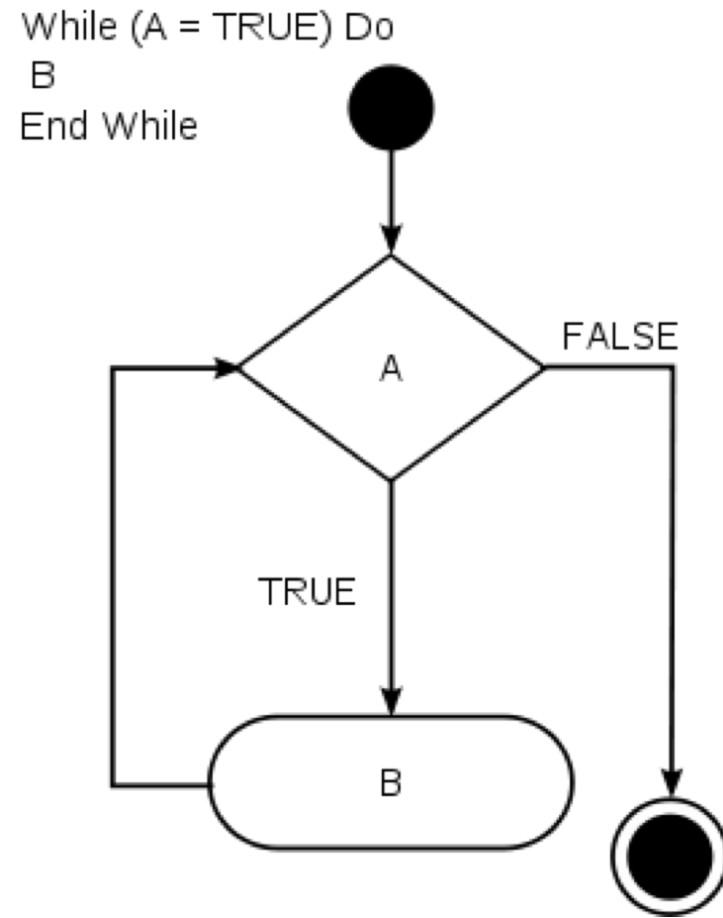
Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

while loop



Source: https://en.wikipedia.org/wiki/While_loop

while loop

```
4  int main() {  
5  
6      int sum = 0, i = 1;  
7  
8      while (i <= 10) {  
9          sum += i;  
10         i++;  
11     }  
12     cout << sum << "\n";  
13 }
```

What happened?

```
4  int main() {  
5  
6      int sum = 0, i = 1;  
7  
8      while (i <= 10) {  
9          sum += i;  
10     }  
11     cout << sum << "\n";  
12 }
```

while loop

```
4  int main() {  
5  
6      int answer = 7, guess = -1;  
7  
8      while (guess != answer) {  
9          cout << "Please enter a number: ";  
10         cin >> guess;  
11     }  
12     cout << "Correct\n";  
13 }
```

```
Please enter a number: 5  
Please enter a number: 6  
Please enter a number: 7  
Correct  
Program ended with exit code: 0
```

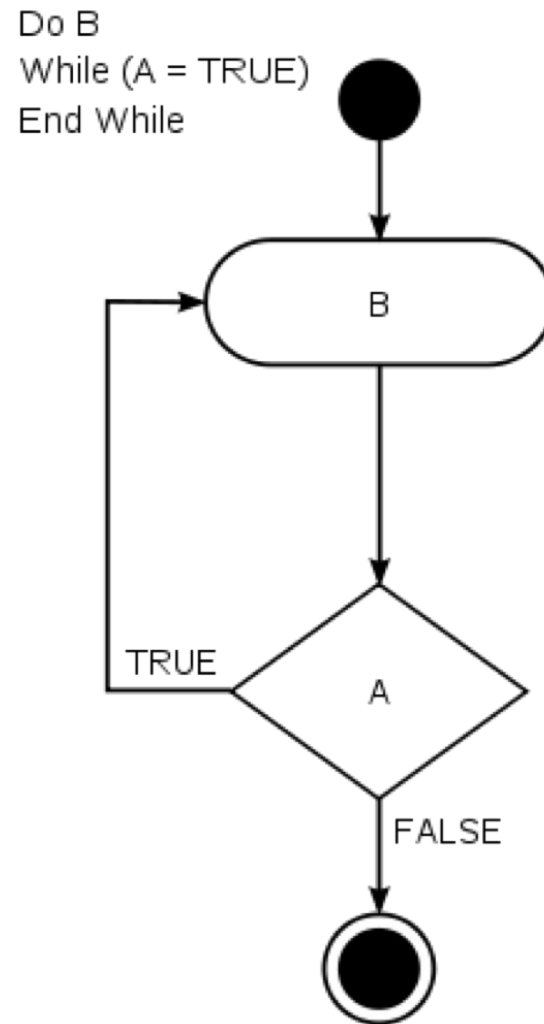
Practice

- (Practice 1) Assume that the initial value of i is 0. Rewrite "Adding number from 1 to 10".
- (Practice 2) Calculate x^y by using while loop.

Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

do ... while



Source: https://en.wikipedia.org/wiki/Do_while_loop

do ... while

```
4  int main() {  
5  
6      int answer = 7, guess = -1;  
7  
8      do {  
9          cout << "Please enter a number: ";  
10         cin >> guess;  
11     } while (guess != answer);  
12     cout << "Correct\n";  
13 }
```

Practice

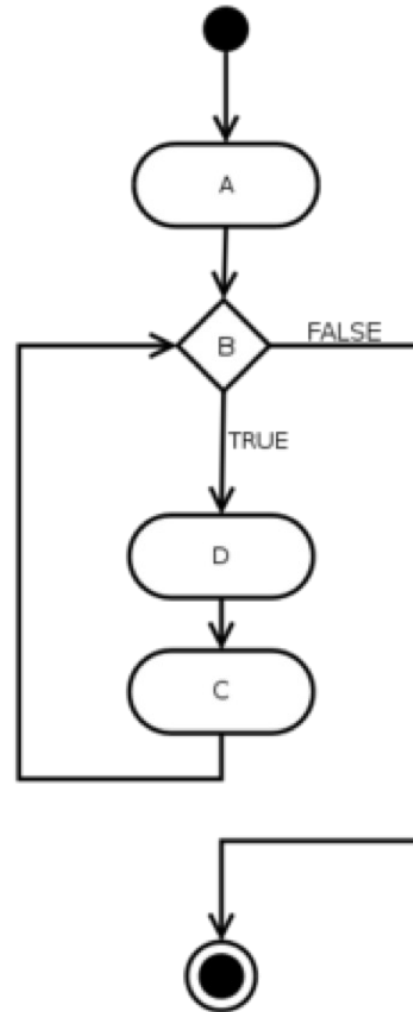
- (Practice 1) Rewrite 'Adding number from 1 to 10' by using do-while.
- (Practice 2) Calculate x^y by using do ... while loop.

Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

for loop

```
for(A;B;C)  
D;
```



(A) initialization (loop variant)

(B) the condition

(C) the advancement to the next iteration

Source: https://en.wikipedia.org/wiki/For_loop

for loop

老師的建議：

感覺學習有點辛苦同學請從
追蹤程式執行過程開始

```
4  int main() {  
5  
6      int sum = 0;  
7      for (int i = 1; i <= 3; i++) {  
8          sum += i;  
9      }  
10     cout << sum << "\n";  
11 }
```

- (1) $i = 1$ ($\text{sum} = 0, i = 1$)
- (2) $(1 \leq 3) \rightarrow \text{true}$
- (3) $\text{sum} = 0 + 1$ ($\text{sum} = 1, i = 1$)
- (4) $i++$ ($\text{sum} = 1, i = 2$)
- (5) $(2 \leq 3) \rightarrow \text{true}$
- (6) $\text{sum} = 1 + 2$ ($\text{sum} = 3, i = 2$)
- (7) $i++$ ($\text{sum} = 3, i = 3$)
- (8) $(3 \leq 3) \rightarrow \text{true}$
- (9) $\text{sum} = 3 + 3$ ($\text{sum} = 6, i = 3$)
- (10) $i++$ ($\text{sum} = 6, i = 4$)
- (11) $(4 \leq 3) \rightarrow \text{false}$
- (12) print 6

Trace the code

```
4  int main() {  
5  
6      int sum = 0;  
7      for (int i = 2; i <= 8; i = i + 2) {  
8          sum = sum + i;  
9      }  
10     cout << sum << "\n";  
11 }
```


for loop

- Infinite loop

```
4  int main() {  
5  
6      int sum = 0;  
7      for (;;) {  
8          if (sum == 1) {  
9              break;  
10         }  
11     }  
12     cout << "End\n";  
13 }
```

for loop (std::setw)

- <https://www.runoob.com/w3cnote/cpp-func-setw.html>

```
1  #include<iostream>
2  #include<iomanip>
3  using namespace std;
4
5  int main() {
6
7      for (int i = 1; i <=9; i++) {
8          for (int j = 1; j <=9; j++) {
9              cout << i << "*" << j << "="
10                 << setw(2) << (i*j) << "\t";
11          }
12          cout << endl;
13      }
14 }
```

for loop (std::setw)

```
1  #include<iostream>
2  #include<iomanip>
3  using namespace std;
4
5  int main() {
6
7      const int maxn = 10;
8      cout << setw(8) << "integer" << setw(8) << "sum"
9           << setw(20) << "factorial" << endl;
10
11     for (int i {1}, sum {}, factorial{1}; i <= maxn; i++) {
12         sum += i;
13         factorial *= i;
14         cout << setw(8) << i << setw(8) << sum
15              << setw(20) << factorial << endl;
16     }
17 }
```

Range-based for (Modern C++)

- `for (range_declaration: range_expression)`

```
4  int main() {  
5  
6      int total {};  
7      for (auto x: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}) {  
8          total += x;  
9      }  
10     cout << total << endl;  
11 }
```

```
[lincyu@Chih-YudeMacBook-Pro Chapter5 % clang++ -std=c++11 5_16_ModernCPP_for.cpp
```

```
[lincyu@Chih-YudeMacBook-Pro Chapter5 % ./a.out
```

```
55
```

```
[lincyu@Chih-YudeMacBook-Pro Chapter5 %
```

Range-based for (Modern C++)

- `for (range_declaration: range_expression)`

```
1  #include<iostream>
2  using namespace std;
3
4  int main() {
5
6      int total {};
7      int iarray[] {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};
8      for (int x: iarray) {
9          total += x;
10     }
11     cout << total << endl;
12 }
```

Practice

- (Practice 1) Calculate x^y by using for loop.
- (Practice 2) Rewrite "Guess Number" by using for loop.
- (Practice 3) Given 12345, output 1, 2, 3, 4, 5 (Given one integer number, output 5 integer numbers.)

```
int number = 67395;  
  
int n5, n4, n3, n2, n1;  
  
n5 = number/10000;  
n4 = (number%10000)/1000;
```

Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

Nested Loops

```
6      int sum = 0;
7      for (int i = 0; i < 2; i++) {
8          for (int j = 0; j < 2; j++) {
9              sum = sum + i + j;
10         }
11     }
12     cout << sum << "\n";
```

(1) $i = 0$ ($\text{sum} = 0, i = 0$)
(2) $(i < 2) \rightarrow \text{true}$
(3) $j = 0$ ($\text{sum} = 0, i = 0, j = 0$)
(4) $(j < 2) \rightarrow \text{true}$
(5) $\text{sum} = 0 + 0 + 0$ ($\text{sum} = 0, i = 0, j = 0$)
(6) $j++$ ($\text{sum} = 0, i = 0, j = 1$)
(7) $(j < 2) \rightarrow \text{true}$
(8) $\text{sum} = 0 + 0 + 1$ ($\text{sum} = 1, i = 0, j = 1$)
(9) $j++$ ($\text{sum} = 1, i = 0, j = 2$)
(10) $(j < 2) \rightarrow \text{false}$
(11) $i++$ ($\text{sum} = 1, i = 1, j = 2$)
(12) $(i < 2) \rightarrow \text{true}$
(13) $j = 0$ ($\text{sum} = 1, i = 1, j = 0$)
(14) $(j < 2) \rightarrow \text{true}$
(15) $\text{sum} = 1 + 1 + 0$ ($\text{sum} = 2, i = 1, j = 0$)
(16) $j++$ ($\text{sum} = 2, i = 1, j = 1$)
(17) $(j < 2) \rightarrow \text{true}$
(18) $\text{sum} = 2 + 1 + 1$ ($\text{sum} = 4, i = 1, j = 1$)
(19) $j++$ ($\text{sum} = 4, i = 1, j = 2$)
(20) $(j < 2) \rightarrow \text{false}$
(21) $i++$ ($\text{sum} = 4, i = 2, j = 2$)
(22) $(i < 2) \rightarrow \text{false}$

Trace the code (1)

```
4  int main() {
5
6      int sum = 0, i = 0, j = 0;
7      for (i = 1; i <= 2; i++) {
8          for (int j = 2; j >= 1; j--) {
9              sum = sum + i + j;
10         }
11     }
12     cout << sum << ", " << i << ", " << j << "\n";
13 }
```

老師的提醒：

int j = 2; 的 int 是一個要注意的陷阱

老師的建議：

不要使用相同的變數名稱

Trace the code (2)

```
4  int main() {  
5  
6      int sum = 0, i = 0, j = 0;  
7      for (i = 5; i <= 8; i = i + 2) {  
8          for (int j = 9; j >= 5; j = j - 3) {  
9              sum = sum + i - j;  
10         }  
11     }  
12     cout << sum << ", " << i << ", " << j << "\n";  
13 }
```

Multiplication Table

```
4  int main() {  
5  
6      for (int i = 1; i <= 9; i++) {  
7          for (int j = 1; j <= 9; j++) {  
8              cout << i << '*' << j << '=' << (i*j) << '\t';  
9          }  
10         cout << "\n";  
11     }  
12 }
```

1*1=1	1*2=2	1*3=3	1*4=4	1*5=5	1*6=6	1*7=7	1*8=8	1*9=9
2*1=2	2*2=4	2*3=6	2*4=8	2*5=10	2*6=12	2*7=14	2*8=16	2*9=18
3*1=3	3*2=6	3*3=9	3*4=12	3*5=15	3*6=18	3*7=21	3*8=24	3*9=27
4*1=4	4*2=8	4*3=12	4*4=16	4*5=20	4*6=24	4*7=28	4*8=32	4*9=36
5*1=5	5*2=10	5*3=15	5*4=20	5*5=25	5*6=30	5*7=35	5*8=40	5*9=45
6*1=6	6*2=12	6*3=18	6*4=24	6*5=30	6*6=36	6*7=42	6*8=48	6*9=54
7*1=7	7*2=14	7*3=21	7*4=28	7*5=35	7*6=42	7*7=49	7*8=56	7*9=63
8*1=8	8*2=16	8*3=24	8*4=32	8*5=40	8*6=48	8*7=56	8*8=64	8*9=72
9*1=9	9*2=18	9*3=27	9*4=36	9*5=45	9*6=54	9*7=63	9*8=72	9*9=81

Program ended with exit code: 0

Practice

- (Practice 1)

1*1=1	2*1=2	3*1=3	4*1=4	5*1=5	6*1=6	7*1=7	8*1=8	9*1=9
1*2=2	2*2=4	3*2=6	4*2=8	5*2=10	6*2=12	7*2=14	8*2=16	9*2=18
1*3=3	2*3=6	3*3=9	4*3=12	5*3=15	6*3=18	7*3=21	8*3=24	9*3=27
1*4=4	2*4=8	3*4=12	4*4=16	5*4=20	6*4=24	7*4=28	8*4=32	9*4=36
1*5=5	2*5=10	3*5=15	4*5=20	5*5=25	6*5=30	7*5=35	8*5=40	9*5=45
1*6=6	2*6=12	3*6=18	4*6=24	5*6=30	6*6=36	7*6=42	8*6=48	9*6=54
1*7=7	2*7=14	3*7=21	4*7=28	5*7=35	6*7=42	7*7=49	8*7=56	9*7=63
1*8=8	2*8=16	3*8=24	4*8=32	5*8=40	6*8=48	7*8=56	8*8=64	9*8=72
1*9=9	2*9=18	3*9=27	4*9=36	5*9=45	6*9=54	7*9=63	8*9=72	9*9=81

Program ended with exit code: 0|

Practice

- (Practice 2) Print

1

12

123

1234

12345

- (Practice 3) Print

54321

5432

543

54

5

Practice

- (Practice 4) Print

```
12345
1234
123
12
1
```

- (Practice 5) Print

```
  *
 * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * * *
```

Outline

- while loop
- do ... while
- for loop
- nested loops
- break / continue

break / continue

- break
 - **terminate** a for, while, or do-while loop
- continue
 - **skips the current iteration** of a for, while , or do-while loop

老師的提醒：

Nested loop 中的 break / continue 要熟練

break

```
4  int main() {  
5  
6      int sum = 0;  
7      for (int i = 1; i <= 10; i++) {  
8          sum += i;  
9          if (i % 2 == 0) {  
10             break;  
11         }  
12         cout << i << "\n";  
13     }  
14     cout << sum << "\n";  
15 }
```

continue

```
4  int main() {  
5  
6      int sum = 0;  
7      for (int i = 1; i <= 10; i++) {  
8          sum += i;  
9          if (i % 2 == 0) {  
10             continue;  
11         }  
12         cout << i << "\n";  
13     }  
14     cout << sum << "\n";  
15 }
```

break (nested loop)

```
6      for (int i = 1; i <= 10; i++) {  
7          for (int j = 0; j <= 10; j++) {  
8              if (j > 2) {  
9                  break;  
10             }  
11             cout << i << ", " << j << "\n";  
12         }  
13     }
```

老師的提醒：
要留意的是“終止”哪一個迴圈

continue (nested loop)

```
6      for (int i = 1; i <= 10; i++) {
7          if (i % 2 == 1) {
8              continue;
9          }
10         for (int j = 0; j <=10; j++) {
11             if (j % 2 == 0) {
12                 continue;
13             }
14             cout << i << ", " << j << "\n";
15         }
16     }
```

break / continue

```
6      for (int i = 1; i <= 5; i++) {  
7          if (i % 2 == 1) {  
8              continue;  
9          }  
10         for (int j = 0; j <= 5; j++) {  
11             if (j >= 3) {  
12                 break;  
13             }  
14             cout << i << ", " << j << "\n";  
15         }  
16     }
```

break / continue

```
6      for (int i = 1; i <= 5; i++) {  
7          if (i % 2 == 1) {  
8              continue;  
9          }  
10         for (int j = 0; j <= 5; j++) {  
11             if (j >= 3) {  
12                 break;  
13             }  
14             cout << i << ", " << j << "\n";  
15         }  
16     }
```

break / continue

```
6      for (int i = 1; i <= 5; i++) {  
7          if (i % 2 == 1) {  
8              continue;  
9          }  
10         for (int j = 0; j <= 5; j++) {  
11             if (j >= 3) {  
12                 break;  
13             }  
14             cout << i << ", " << j << "\n";  
15         }  
16     }
```